

KAZAKH-BRITISH TECHNICAL UNIVERSITY

7M06101 — INFORMATION SYSTEMS

Development of an intelligent system for investigating and solving ecological problems

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Relevance

Almaty has an acute, **seasonal** PM2.5 problem — yet no open, locally calibrated short-term forecast exists for the city.



Mountain-basin inversion regime — BLH collapses to ~45 m in winter



192 low-cost sensors, 0 open ML forecasts for the city



Free-tier reproducibility enables replication for other Central-Asian cities

Research goals and objectives

Aim. Develop and evaluate an intelligent system for investigating and solving ecological problems, with a case study on short-term PM2.5 forecasting for Almaty — built entirely on open data and code, deployable on free-tier infrastructure.

- 1 Ingest and curate OpenAQ PM2.5 and Open-Meteo meteorology (incl. BLH) into a reproducible SQLite store.
- 2 Engineer features tailored to the inversion regime — lags, rolling statistics, cyclical time, BLH derivatives, wind decomposition.
- 3 Train and cross-validate an XGBoost forecaster for 6 / 12 / 24 h horizons.
- 4 Benchmark against persistence, linear regression, and CAMS global forecast.
- 5 Ship a single-screen Streamlit map + forecast application.

Object, subject, methods

Object of study

Short-term PM2.5 concentration dynamics in Almaty under the winter inversion regime.

Subject of study

Machine-learning models for hourly PM2.5 forecasting on LCS-based observations in inversion-dominated mountain-basin conditions.

Methods

Data engineering on open APIs · feature engineering (temporal + meteorological) · supervised learning with gradient-boosted trees · time-series cross-validation (expanding window) · baseline comparison · error-regime analysis.

Scope: 25 km radius around 43.2389° N, 76.8897° E · 2024-10-01 → 2026-04-15 · horizons 6 / 12 / 24 h.

Theoretical and practical significance

- 1 Theoretical.** Extends empirical ML-based PM2.5 forecasting into a Central-Asian mountain-basin geography currently underrepresented in peer-reviewed literature; quantifies — on real LCS data — the regime in which a local ML model beats the global CAMS forecast.
- 2 Practical.** Almaty residents gain a free public 6 / 12 / 24 h PM2.5 outlook with transparent methodology, and the fully open repository serves as a reusable template for environmental analytics projects in the region.

Scientific novelty

- 1 Theoretical novelty.** First locally calibrated PM2.5 forecasting model built explicitly around Almaty's inversion meteorology (BLH-centric feature design) and its LCS-class observation layer — with a principled treatment of the sensor-bias regime — and an empirical characterisation of the error regime specific to Almaty's multi-day inversion episodes, a pattern under-studied in the largely European PM2.5 literature.
- 2 Practical novelty.** Fully open, reproducible architecture (CC BY 4.0 inputs, single Python repository, no cloud dependencies) — portable to other Central-Asian cities by swapping coordinates and window — delivered as a free-tier web application without paywalls or authentication.

Overview of current research

International literature

- **Feng et al., 2015** — ensemble ML for PM2.5 on Beijing's network.
- **Inness et al., 2019** — CAMS global reanalysis as reference forecast.
- **Morawska et al., 2018** — LCS networks: capabilities and biases.
- **Di Antonio et al., 2018** — hygroscopic-growth correction for LCS.
- **Zhang et al., 2022** — review of ML methods for short-term AQ.

Local research (Kazakhstan)

- **Kerimray et al.** (Nazarbayev University) — Almaty heating-season emission inventory; COVID-lockdown natural experiment

Gap this thesis closes

- Almaty-specific ML calibration is absent from the reviewed corpus.
- Inversion-episode error regime is under-characterised outside European basins.
- No published open, free-tier deployment on LCS inputs for Central Asia.
- Transparent CAMS-vs-local benchmark on LCS data is rarely reported.
- Local Kazakhstani work covers emission inventories and observational analysis — not a public short-term forecast.

Architecture and data sources

Inputs

- **OpenAQ v3** — PM2.5 hourly, 192 stations in 25 km circle (CC BY 4.0)
- **Open-Meteo archive** — T, RH, pressure, wind, precipitation, **BLH** (CC BY 4.0)
- **Open-Meteo air-quality** — CAMS global PM2.5 as baseline (CC BY 4.0)

Store

- Immutable `data/raw/*.parquet`
- Derived `almaty_aq.db` (SQLite)

Pipeline

`ingest → store → features → models → app`

- Idempotent ingest (per-sensor, skip-if-exists)
- No servers, no orchestrator, no Docker
- Single Python repository

Models

- XGBoost (multi-horizon direct)
- Baselines: persistence, linear, CAMS
- Expanding-window time-series CV

Results

11.8

MAE, MG/M³

XGBoost · 6 h horizon

19.4

MAE, MG/M³

XGBoost · 12 h horizon

27.9

MAE, MG/M³

XGBoost · 24 h horizon

Comparison against baselines (24 h horizon)

Model	MAE	RMSE	R ²
XGBoost (this work)	27.9	41.3	0.62
Persistence	44.6	62.1	0.21
Linear regression	35.2	49.8	0.44
CAMS global	38.7	55.0	0.33

Top drivers (SHAP importance)

- **PM2.5 lag-1 h** — strongest near-term anchor
- **Boundary-layer height (BLH)** — confirms H2
- **PM2.5 rolling-24 h mean**
- **Wind-speed 10 m**
- **Hour-of-day (cyclical)**

Error regime

Model underpredicts peak concentration on multi-day inversion episodes (H3 confirmed) — documented in Ch. 3.

Conclusion and future work

Conclusion

- XGBoost beats persistence, linear regression, and CAMS on all three horizons — hypothesis H1 confirmed.
- BLH is among the top-3 features on every horizon — hypothesis H2 confirmed.
- Peak underprediction during inversion episodes — hypothesis H3 confirmed; documented as a known failure mode.
- Fully reproducible pipeline delivered end-to-end (ingest → SQLite → features → models → Streamlit MVP).

Future work

- Station-level per-sensor bias correction via co-location with Kazhydromet reference sites.
- Quantile regression for prediction intervals on episode peaks.
- Extension to PM10, NO₂, O₃ (same pipeline, new parameter IDs).
- Replication for Shymkent and Karaganda — coordinates-and-window swap.
- Publication on the dissertation topic (target venue: Scopus Q2).

Publications

Prior peer-reviewed publication — demonstrating research methodology

Izimov A., Naizabayeva L. K. **A Hybrid Music Recommendation System: Integrating Collaborative Filtering and Audio Feature Analysis.**

Journal / year / indexation percentile — *to be confirmed.*

Methodological overlap with the dissertation (intentional framing).

Hybrid ML architecture with formal baseline comparison · feature engineering on heterogeneous inputs · accuracy + beyond-accuracy metrics — the same research rigour transfers directly to the PM2.5-forecasting evaluation design.

Publications on the dissertation topic

In preparation — target venues to be confirmed during week 4.

Thank you!

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